Power flags (ERC drivers for rails that enter through connectors)

VP_IN VN_IN GND

Connector and rail filter

VP_IN 1
GND 2
VN_IN 3
OUT 4
INN 5
INP 6
J1
HVOA_IF

R1
10

C1
100u 63V

C2
100n 100V

R2
10

C3
100u 63V

C4
100n 100V

J1 mates with the 6-pin socket on the preamp board.

References (red LEDs ~1.7 V)

D101
red 3mm
REFP VP

R101
39k

C101
10u 63V

D102
red 3mm
VN REFN

R102
39k

C102
10u 63V

Current mirror 2N5210

M1 Q5
2N5210

M2 Q6
2N5210

R107
1k

R108
1k

Input pair 2N5087 with bootstrapped MPSA92 cascode

REFP Q103
MPSA92

R103
3k0 / A2: 820

R104
1k / A2: 470

R105
1k / A2: 470

INN Q1
2N5087

INP Q2
2N5087

D103
1N5232B 5V6
TAIL CASB

C103
100n

REFN Q104
MPSA42

R106
10k / A2: 3k3

CASB Q3
MPSA92

CASB Q4
MPSA92

D103 holds the cascode bases 5.6 V below TAIL: Vce(Q1,Q2) = 4.7 V at any common mode.

VAS: MPSA42 follower, 2N5210 VAS, MPSA42 cascode, MPSA92 load

D104
1N5226B 3V3
VCR VN

R109
33k

C104
10u 16V

M2 Q7
MPSA42

R110
22k

V7 Q8
2N5210

R111
100

EB Q15
2N5210

VCR Q9
MPSA42

REFP Q105
MPSA92

R112
270

CC1
68p / A2: 150p

CC: C0G ceramic or silver mica, >= 100 V.

Output stage (2 mA idle, 11 mA limit)

D105
1N4148
B1 VASH

D106
1N4148
B2 B1

R113
100

VASH Q11
MPSA42

VASL Q12
MPSA92

R114
56

R115
56

E11 Q13
MPSA42

E12 Q14
MPSA92

All TO-92 parts are E-B-C pinout (2N5087, 2N5210, MPSA42, MPSA92).
Match Q1/Q2 (Vbe 2 mV, hFE 10 %) and Q5/Q6; glue each pair together. D105/D106 touch Q11/Q12.
Netlist identical to bass-amp/sim/hvoa.sub (checked by gen/verify_hvoa.py).
Build twice: A1 (instrument input) and A2 (FX return). Parts marked "A1 / A2" differ.

Sheet: /
File: hvoa_module.kicad_sch

Title: HVOA-1 +/-40 V nested-cascode discrete op-amp module

Size: A2	Date:	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1